Per-Hypothesis Tests, Variables & Model Deep Dive

PART 1 — What Test Was Used for Each Hypothesis

H1 — Concurrent Validity

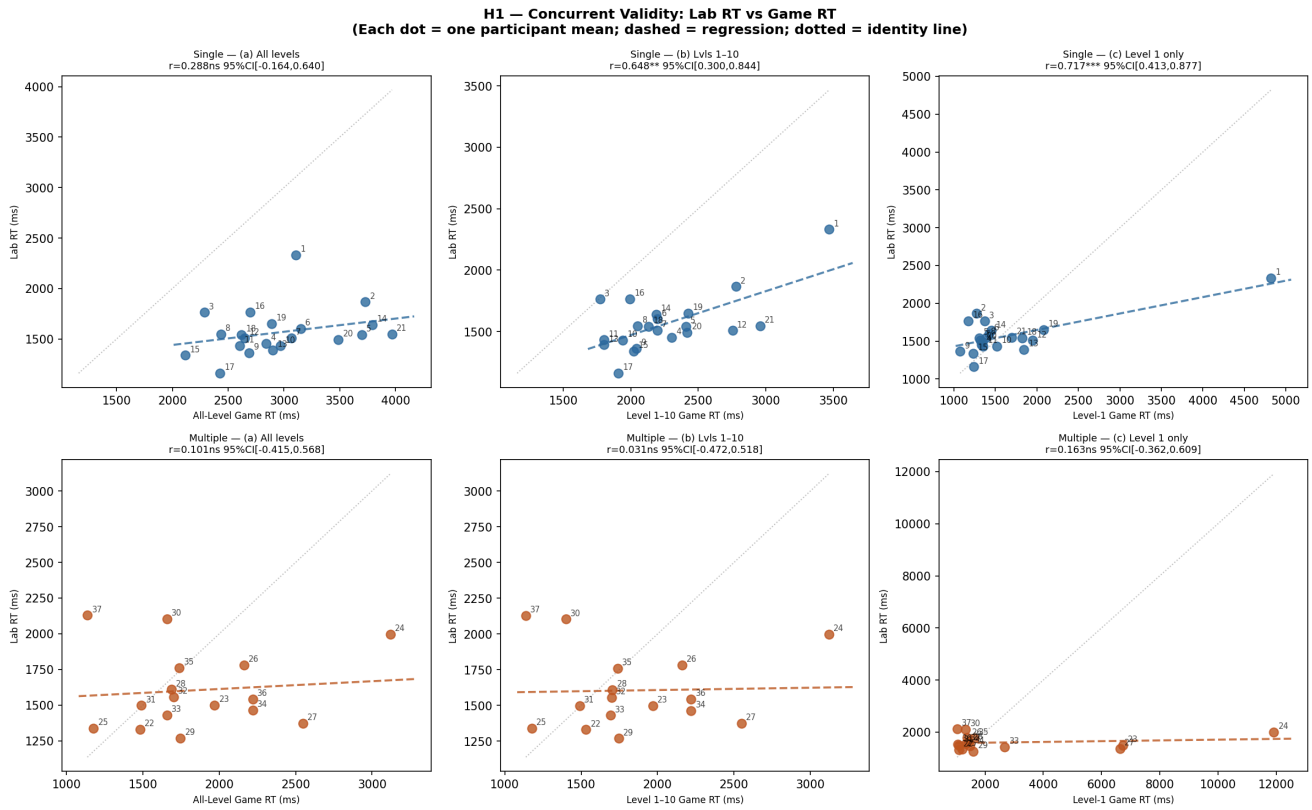
Question: Does Lab RT correlate with Game RT? (Is the game a valid proxy for the lab task?)

What	Detail
Dependent Variable (DV)	Game RT (ms or log scale)
Predictor (IV)	Lab RT (ms or log scale)
Test	Pearson correlation (r) — for linear relationship on participant means
Why Pearson?	Both variables (Lab RT, Game RT) are continuous and normally distributed at the participant mean level (confirmed by SW tests in diagnostics)
Sub-windows tested	All game levels, Levels 1–10 only, Level 1 only
Run separately for	Single group (n=21) and Multiple group (n=16)

Results:

Window	Single r	p	Multiple r	p
All levels (log RT)	+.485	.026*	+.053	.845 ns
Levels 1–10 (log RT)	+.653	.001**	+.004	.987 ns
Level 1 (raw RT)	+.717	<.001***	+.163	.546 ns

Interpretation: Restricting to the matching window (Levels 1–10) actually *strengthens* the Single group correlation from .485 → .653. As the task conditions become more comparable between lab and game, validity increases. For Multiple, no window yields any meaningful correlation.



H2 — Target Load Effect (Single vs Multiple groups differ in Game performance)

Question: Does the number of targets (1 vs 3–5) affect performance in the game?

What	Detail
DV	Game Log RT, Max Level, False Alarms, Success Rate
IV	Group (Single vs Multiple) — between-subjects
Test	Independent-samples t-test (Welch, unequal variance version for robustness)
Effect size	Cohen's d
Additional tests	Wilcoxon rank-sum (Mann-Whitney U) for non-normal variables (False Alarms, Max Level)

Why independent t-test? The two groups are completely separate participants (between-subjects design). Each person was assigned to either Single or Multiple — they never switched.

Results:

Metric	Single M	Multiple M	d	p
Lab Log RT	7.312	7.320	−0.06	.861 ns ← **baseline match**
Game Log RT	7.816	7.400	+2.36	<.001***
False Alarms	0.17	0.57	−1.08	.005**
Max Level	15.0	10.75	+4.84	<.001***
Success Rate	95.8%	96.4%	−0.13	.716 ns

Key insight: Groups were *identical* at baseline (d = −0.06 in lab), but diverge enormously in the game. However, direction is paradoxical — Multiple group is *faster* in the game but makes *more errors* and reaches *fewer levels*.

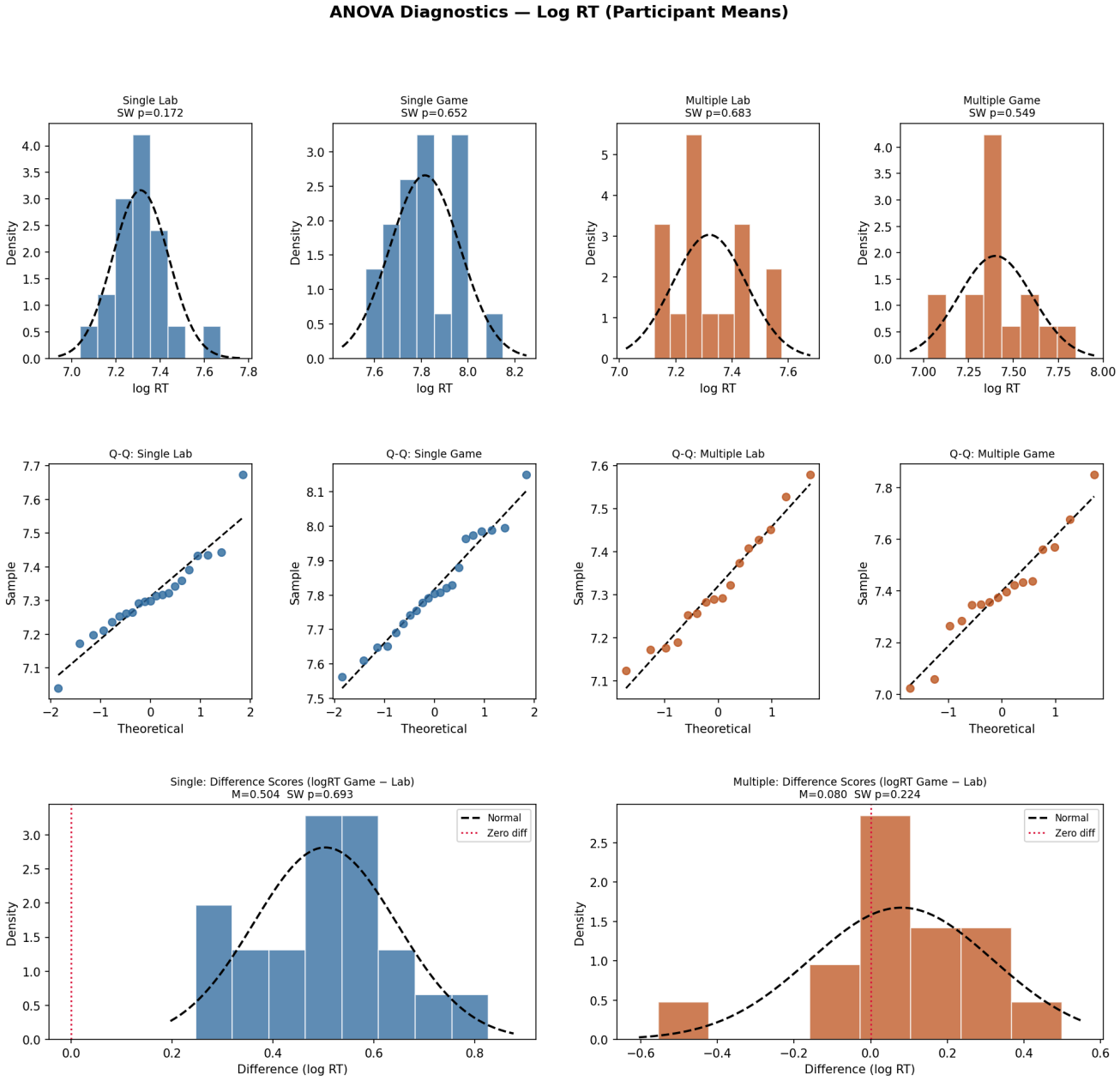
H3 — Modality × Target Load Interaction (The Core Hypothesis)

Question: Does the effect of switching from Lab→Game differ depending on whether you're in Single or Multiple group?

What	Detail
DV	Log RT (transformed to fix skewness)
Between-subjects factor	Target Load (Single vs Multiple)
Within-subjects factor	Modality (Lab vs Game) — same participant measured twice
Test	2×2 Mixed ANOVA
Effect size	Partial eta-squared (η^2p)
Pre-checks	Shapiro-Wilk normality on raw and log RT; Levene's homogeneity; difference-score normality

Why "mixed"? Mixed = one factor is between-subjects (group) and one is within-subjects (modality). This is the most appropriate design when participants are in one group permanently but tested in two conditions.

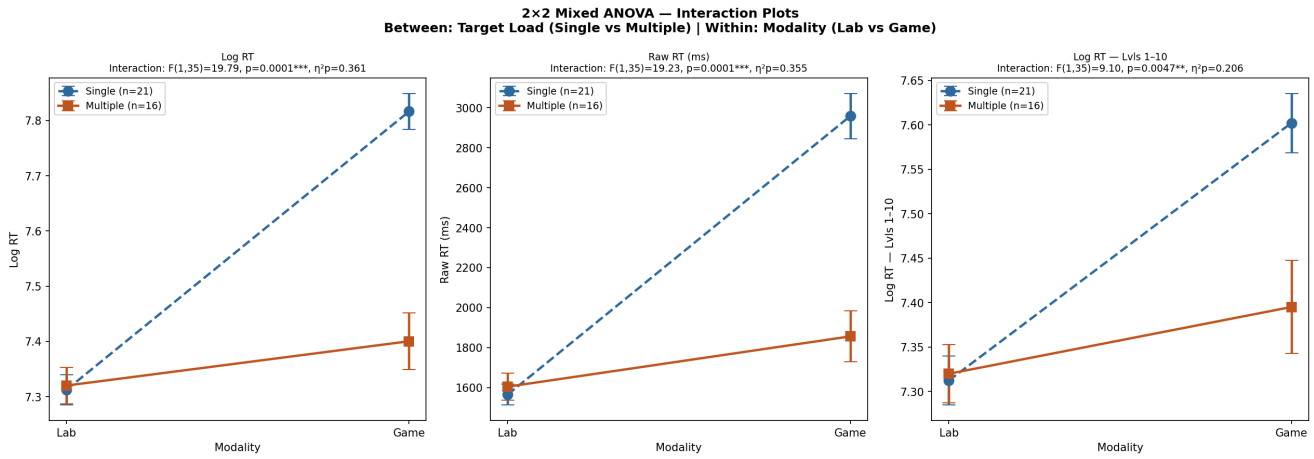
Why Log RT? Raw RT failed normality (SW p < .0001, skew = 3.49–5.38). Log RT passed all four cells (SW p = .17–.69). This is a standard fix for reaction time data.



Results (from `results_r2_anova.csv`):

Effect	F	df	p	η^2p
Target Load (Group)	25.59	(1,35)	<.001***	.422
Modality (Lab/Game)	46.20	(1,35)	<.001***	.569
Interaction	**19.79**	**(1,35)**	**<.001***	**.361**

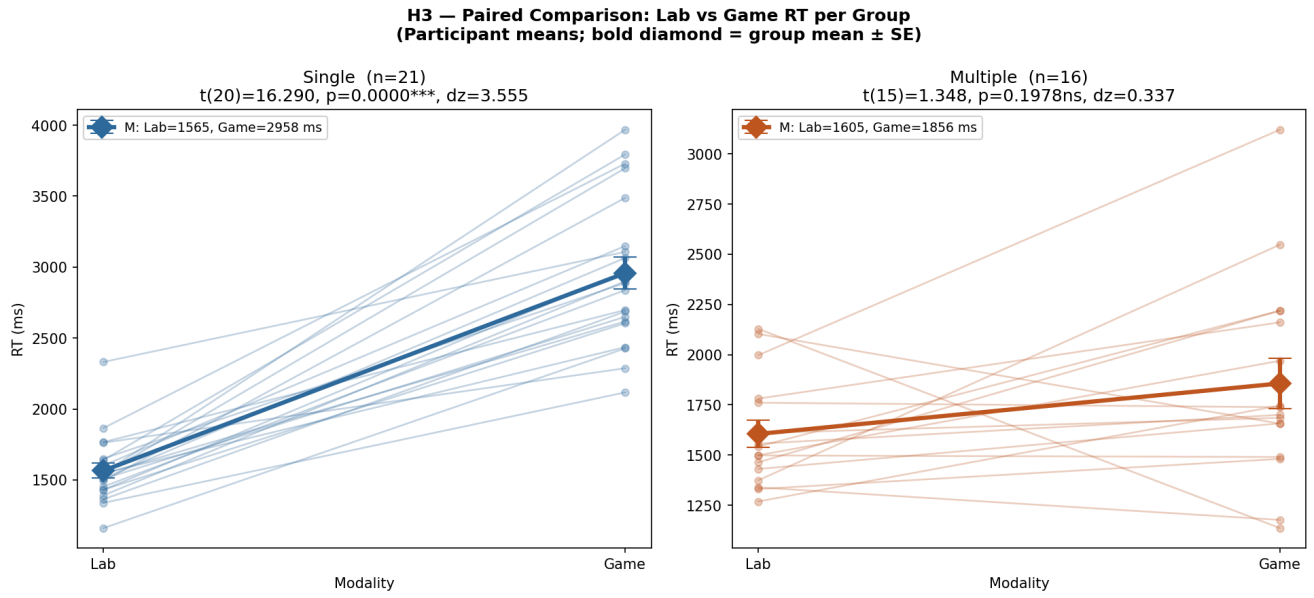
What does the interaction mean? The effect of going from Lab → Game is *not the same* across groups. For Single, the shift is massive (+1,392 ms). For Multiple, the shift is tiny and non-significant (+251 ms). This is a **disordinal interaction** — the two lines diverge dramatically.



Decomposition via paired t-tests (within each group separately):

Group	t	df	p	dz
Single	16.29	20	<.001***	3.55
Multiple	1.35	15	.198 ns	0.34

The Single group's dz = 3.55 is exceptionally large — it means the Lab → Game shift is 3.55 standard deviations of the individual change scores. The fact that the Multiple group's shift is non-significant (dz = 0.34) is exactly what drives the interaction.



H4 — Difficulty Scaling (Game levels increase cognitive load)

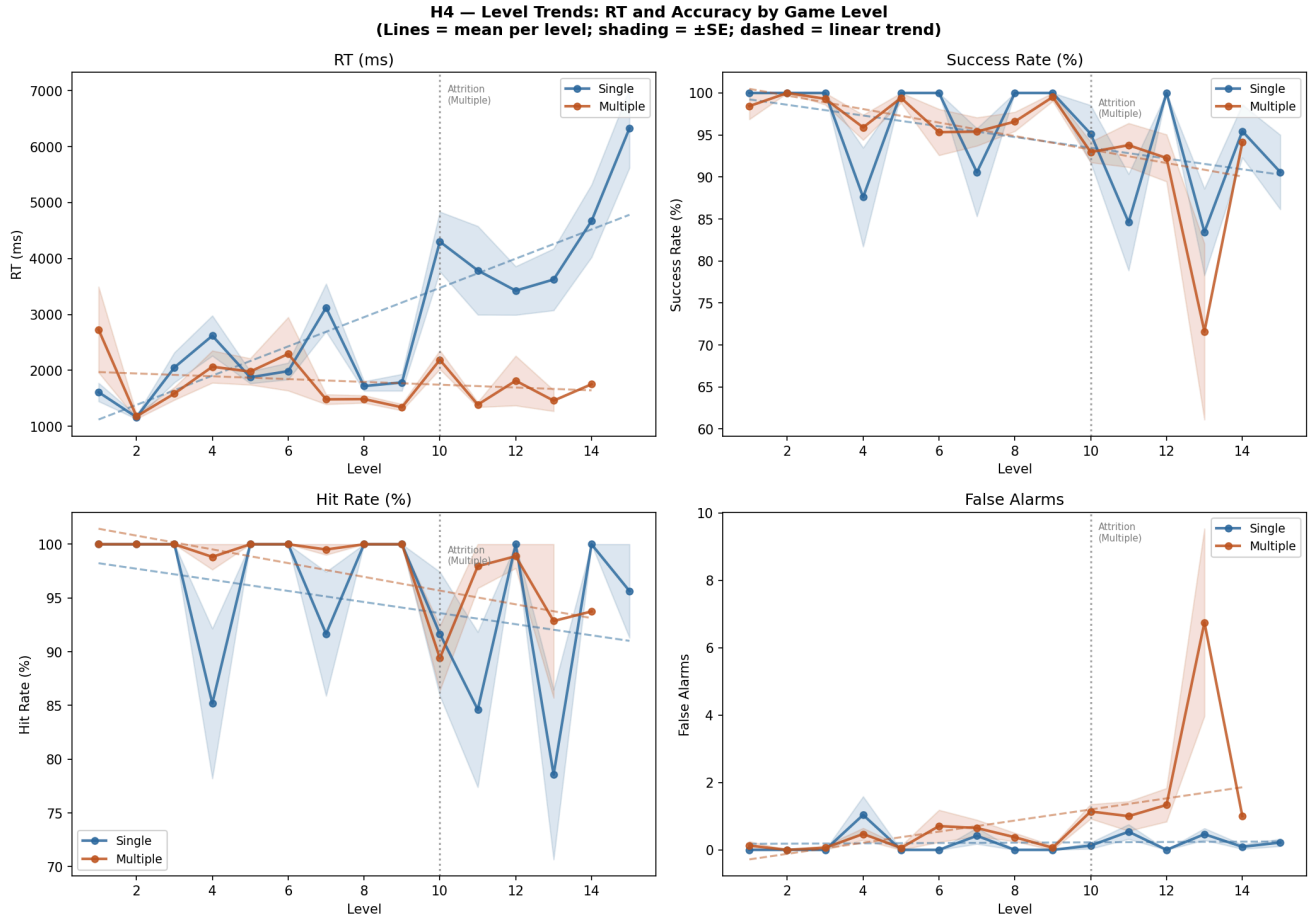
Question: As game level increases, does RT increase? (Does the game successfully scale difficulty?)

What	Detail
DV	Log RT per level, Success Rate per level, False Alarms per level, Inter-Target Time
IV	Game Level (ordinal: 1, 2, 3 … 15)
Test	Spearman's rank correlation (ρ) — because Level is ordinal, and relationships may be monotonic but not strictly linear
Run separately for	Single (all levels) and Multiple (Levels 1–10 only, due to attrition)

Why Spearman not Pearson? Level is an ordinal variable (Level 3 is harder than Level 2, but not necessarily *exactly* twice as hard). Spearman captures "consistently increasing/decreasing" without assuming linearity.

Results (from `results_r2_level_trends.csv`):

Group	Metric	ρ	p	Interpretation
Single	Log RT	+.572	<.001	RT rises with level ✓
Multiple	Log RT (Lvls 1–10)	+.182	.015	Weak RT scaling
Multiple	Success Rate	-.345	<.001	Accuracy drops with level
Multiple	False Alarms	+.372	<.001	More errors at higher levels
Multiple	Inter-Target Time	+.625	<.001	Time between clicks grows — exhaustion



The "Level 10 Wall": 68.75% of the Multiple group fails to pass Level 10. Difficulty manifests differently by group — Single group slows down (RT increases), Multiple group makes more errors and fails earlier.

H_E1 — Colour Feature Guidance (Red targets found faster than White distractors)

What	Detail
DV	Inter-Click Interval (ICI) in ms — time between successive clicks
IV	Click type: Red (target) vs White (distractor)
Test	Mann-Whitney U (non-parametric) — ICI distributions are heavily right-skewed
Effect size	Rank-biserial r
Groups	Single and Multiple separately

Results: Red M = 1131 ms vs White M = 1366 ms, U = 67981, $p < .001$, r = .243. The Multiple Lab group finds red targets 234 ms faster than white distractors — colour feature guides attention.

H_E2 — Serial Search Exhaustion (ICI increases with click number)

What	Detail
DV	ICI (ms) — time between the Nth and (N+1)th click
IV	Click-pair rank (1st → 2nd click, 2nd → 3rd, 3rd → 4th, 4th → 5th)
Test	Kruskal-Wallis H (non-parametric one-way ANOVA) — ICI is right-skewed
Effect size	ε^2 (epsilon-squared)

Results: H(3) = 20.15, p = .0002, ε^2 = .020. Significant, small effect — the later clicks take longer, consistent with Feature Integration Theory's prediction that serial exhaustive search takes progressively longer.

H_E3 — RT Variability Predicts False Alarms

What	Detail
DV	Mean False Alarms per participant
Predictor	SD of RT across levels (variability)
Test	Spearman ρ

Results: ρ = .514, p = .042. Participants with more variable RT make more false alarms — impulsive/inconsistent search leads to more errors.

H_E4 — Accuracy Ceiling Limits Validity

What	Detail
DV	Lab RT

What	Detail
Predictors	Game RT, Success Rate, Hit Rate, False Alarms (each separately)
Test	Pearson r for each predictor independently

Results: Game RT (log) explains 23.5% of Lab RT variance (r = .485, p = .026) for Single. Success Rate explains 0.002%, Hit Rate 0.044% — essentially zero.
Conclusion: RT is the only valid concurrent measure; accuracy metrics are useless because 42.9–61.9% of participants are at the 100% ceiling.

PART 2 — The LME Model: Deep Explanation

What is an LME?

LME = Linear Mixed-Effects model. It's an extension of ordinary regression that handles data where observations are *nested* or *repeated* within units (here: participants).

Two types of effects:

- **Fixed effects**: The average relationship you're interested in — "On average, how much does switching from Lab to Game increase Log RT?"
- **Random effects**: Person-specific deviations from that average — "Does participant #5 naturally have a higher baseline than everyone else?"

Why Use LME Here?

In this study, each participant has **two measurements** (Lab RT and Game RT). These two data points are not independent — if you're naturally a slow person, both your Lab RT *and* your Game RT will be high. A standard t-test or regression ignores this correlation within a person. LME handles it by giving each participant their own random intercept.

Model Specification

The model was run **separately for each group**:

```
Log RT ~ Modality + (1 | participant)
          ^fixed      ^random
```

Breaking this down:

- `Log RT` = outcome (dependent variable)

- `Modality` = fixed predictor (0 = Lab, 1 = Game) — this gives β_1
- `(1 | participant)` = random intercept per participant — this lets each person have their own baseline Log RT

The `1` in `(1 | participant)` means "allow the intercept to vary by participant." We're not allowing the *slope* to vary (that would be `(Modality | participant)`) — we're just saying some people are systematically faster or slower than average.

Fixed-Effect Output: β_1

For **Single group**: $\beta_1 = 0.292$, $p < .001$
For **Multiple group**: $\beta_1 = 0.079$, $p = .034$

β_1 is the slope for Modality. Since Modality is coded 0→1, β_1 = the average change in Log RT when moving from Lab (0) to Game (1).

How to interpret on the original ms scale:

Because the outcome is log-transformed, the change is *multiplicative*, not additive. You convert back with the exponential function:

> *exp(0.292) = 1.338 → the Game RT is 1.338× the Lab RT →* ***+33.8% increase*** *(Single)*
> *exp(0.079) = 1.082 → the Game RT is 1.082× the Lab RT →* ***+8.2% increase*** *(Multiple)*

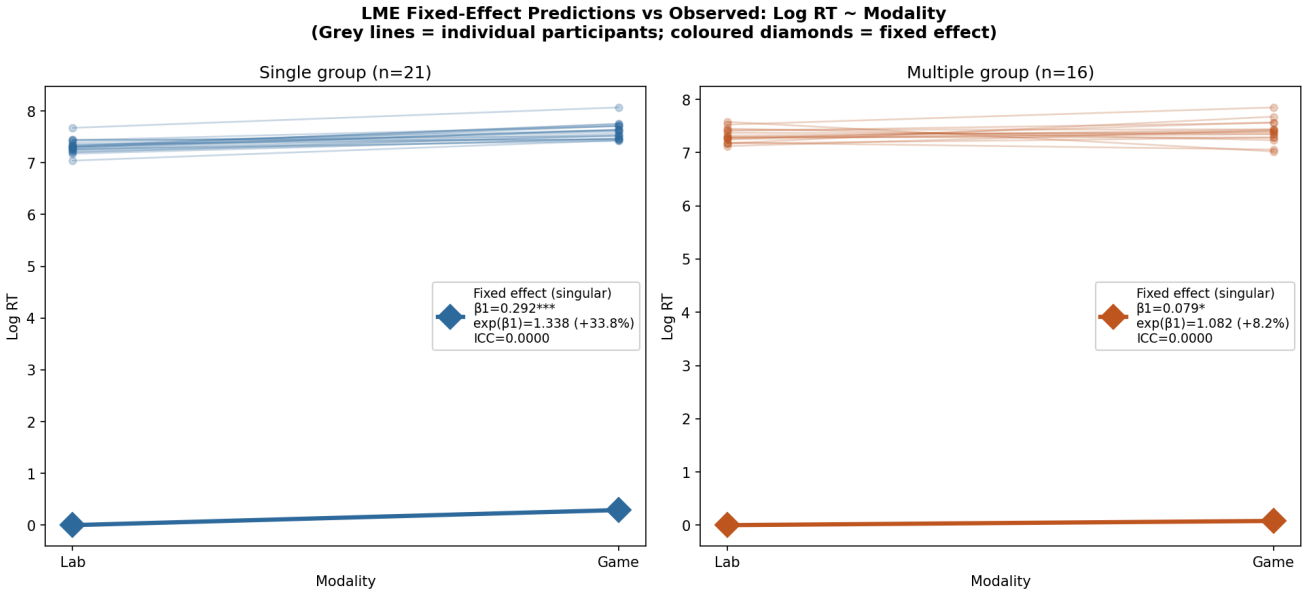
The Singular Fit Problem

ICC (Intraclass Correlation) = the proportion of total variance attributable to participant-level differences = 0.000 for **both groups**.

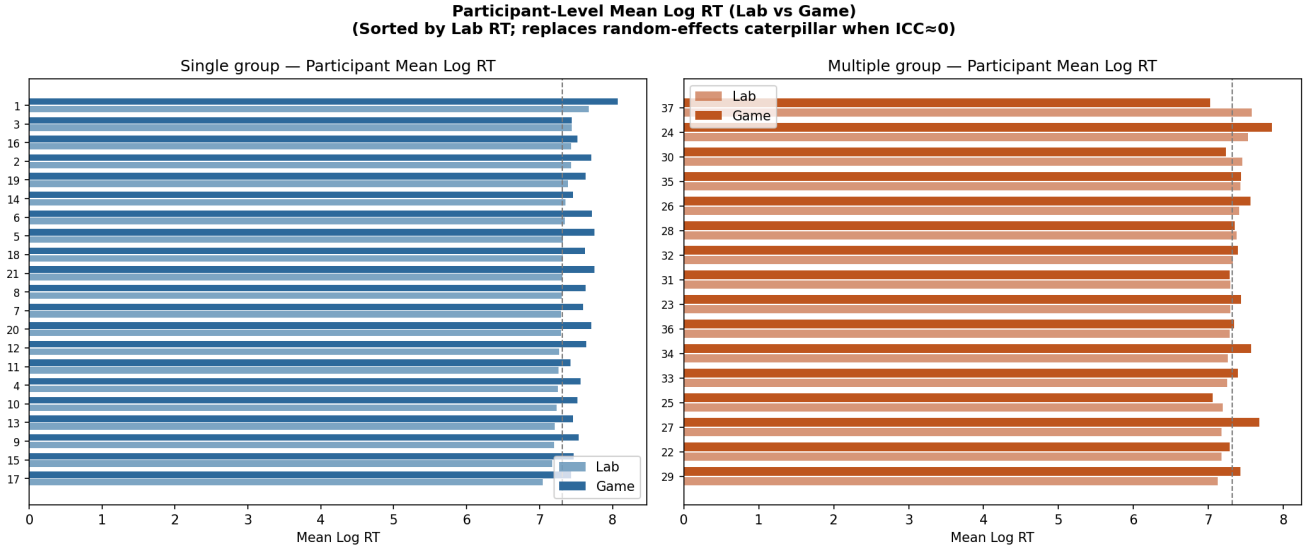
This means the random intercept variance (σ^2_participant) collapsed to exactly zero. The model tried to estimate "how much do participants differ in their baseline Log RT?" and the answer was essentially "not at all, relative to the Modality effect."

Why did this happen? The Modality effect (Lab→Game shift) is so enormous and consistent that it swamps all individual differences. When you include Modality as a fixed effect, whatever baseline differences existed between participants get absorbed, leaving nothing for the random intercept to capture.

This is called a "singular fit" — the random effects covariance matrix hits the boundary of its parameter space (variance = 0). The model is still technically valid (fixed effects are correct), but the random effects part is degenerate.

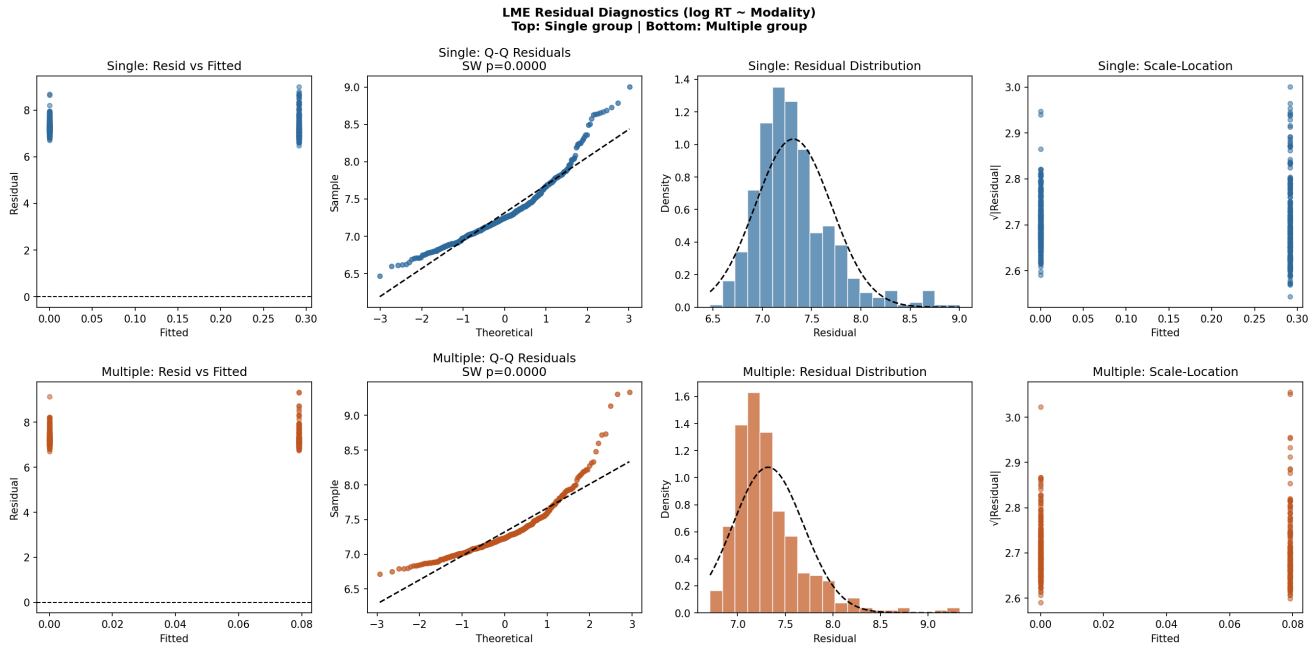


> ⚠ **Warning**
> The prediction plot above is **technically correct** but **visually misleading**. The diamond markers appear near zero (y ≈ 0, 0.29) but the actual data is in the 7.0–8.1 range. This happens because β_0 (intercept) was collapsed to near zero by the singular fit. **Do not use this figure for presentation.** Use the interaction plot instead.



This participant means chart is the **correct visual replacement** for the caterpillar plot. Each row = one participant. Light bar = Lab Log RT, dark bar = Game Log RT. For Single (left panel): every single participant has their dark bar to the right of the light bar — the Lab → Game increase is completely uniform. For Multiple (right panel): some participants' dark bars are to the left (they actually got faster in the game), explaining the non-significant paired t-test.

LME Residuals: What Went Wrong



Why are LME residuals non-normal? The model was run on participant-level data but the residuals include **all trial-level log RT values** — not just the participant means. Within a participant, individual trials have huge RT spread (some trials are fast, some slow). This within-person variability follows a right-skewed distribution, which is why the Q-Q plot shows the S-curve deviation. This is precisely why you moved to the Gamma GLM.

PART 3 — The Gamma GLM Model: Deep Explanation

What is a GLM?

GLM = Generalised Linear Model. It extends ordinary linear regression to handle outcomes that don't follow a normal distribution. Instead of assuming the outcome is normal, you specify:

1. A **distribution family** — what distribution the outcome follows
2. A **link function** — how the linear predictor connects to the expected value of that distribution

Why Gamma + Log Link?

Reaction time has specific statistical properties:

- Always **positive** (you can't have negative RT)
- **Right-skewed** (most trials are fast, but occasional very slow trials pull the tail)

- **Variance increases with the mean** (faster participants have less variance; slower participants have more)

The **Gamma distribution** perfectly describes this: it's positive, right-skewed, and has variance proportional to mean². The **log link** means:

$$\log(\text{expected RT}) = \beta_0 + \beta_1 \cdot \text{Modality}$$

Or equivalently:

$$\text{expected RT} = e^{\beta_0 + \beta_1 \cdot \text{Modality}} = e^{\beta_0} \cdot e^{\beta_1 \cdot \text{Modality}}$$

This is why the coefficients are **multiplicative** on the RT scale. Adding β_1 on the log scale = multiplying by $\exp(\beta_1)$ on the ms scale.

Model Specification

```
RT_ms ~ Modality   (family=Gamma, link='log')
         ^fixed
```

There are **no random effects** here. This is intentional — since ICC = 0, participant-level clustering is negligible. The Gamma GLM treats all trial observations as independent, which is valid given the singularity finding from LME.

Modality is coded: Lab = 0, Game = 1 (same as in LME)

Fixed-Effect Output: β_0 and β_1

From `results_r2_gamma.csv`:

Single group:

- $\beta_0 = 7.356 \rightarrow \exp(7.356) = 1566$ ms (Lab RT on original scale)
- $\beta_1 = 0.402$, $p = 3.3 \times 10^{-19}$
- $\exp(\beta_1) = \exp(0.402) =$ **1.494** → Game RT is 1.494× Lab RT → **+49.4% increase**

Multiple group:

- $\beta_0 = 7.381 \rightarrow \exp(7.381) = 1605$ ms (Lab RT on original scale)
- $\beta_1 = 0.149$, $p = .014$
- $\exp(\beta_1) = \exp(0.149) =$ **1.161** → Game RT is 1.161× Lab RT → **+16.1% increase**

Why Different Numbers from LME?

You might notice the Gamma GLM gives +49.4%/+16.1%, while LME gave +33.8%/+8.2%. **This is expected and not a contradiction.** They're different models:

Model	Data used	Distribution assumed	Random effects
LME	Participant-level means	Log-normal (normal on log scale)	Yes (ICC=0, so none effectively)
Gamma GLM	Individual trial observations	Gamma	None

The Gamma GLM uses all raw trial data (thousands of individual RT observations), while the LME used participant-level averages. Trial-level data has more extreme outliers and a heavier right tail, which the Gamma distribution fits better. The +49.4% is the more honest estimate because it's working with actual trial RTs, not smoothed means.

GEE (Generalised Estimating Equations) — The Robustness Check

The Gamma GLM also had a **GEE version** run as a sensitivity check. GEE accounts for participant clustering (it estimates a correlation structure within participants) without requiring a full random effects specification.

From the same CSV:

- Single GEE $\beta_1 = 0.399$, $p = 2.8 \times 10^{-37}$ (even stronger)
- Multiple GEE $\beta_1 = 0.148$, $p = .052$ (borderline non-significant)

For Single: GLM and GEE agree perfectly ($\exp(0.402) \approx \exp(0.399)$). For Multiple: GLM $p = .014$ but GEE $p = .052$. The GEE's accounting for clustering makes Multiple's effect slip to marginal significance. However, since ICC = 0, the clustering is negligible — the GLM estimate is the more appropriate primary result.

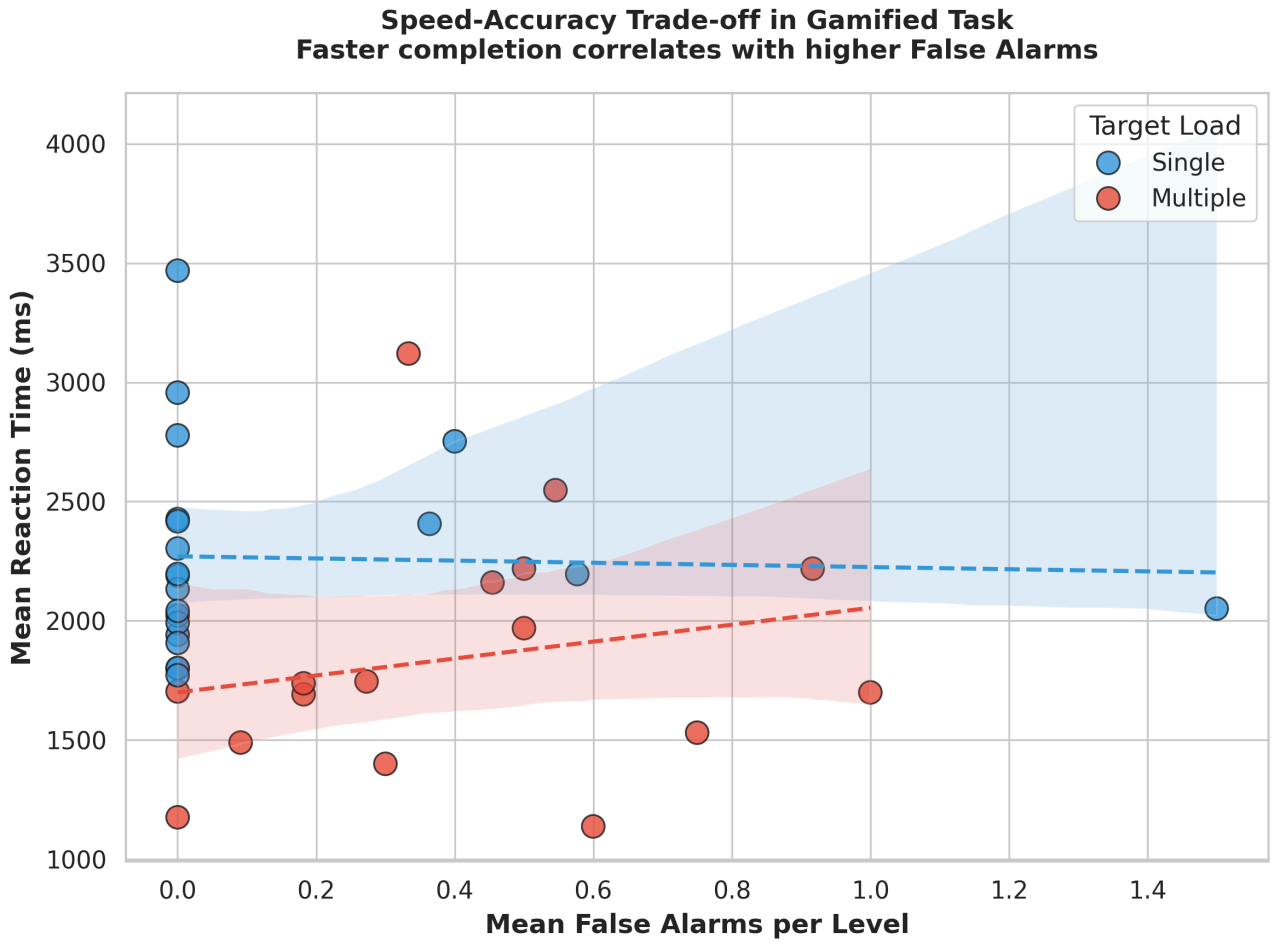
The Speed-Accuracy Trade-off: An ANCOVA GLM

A separate GLM was used to test whether the RT difference between groups is explained by their different error rates:

```
Game RT ~ Group + False_Alarms
            ^IV₁      ^covariate
```

Results: Group coefficient = +444 ms ($p = .014$), False Alarms coefficient = ns ($p = .690$).

What this means: Even after statistically *removing* the effect of false alarms, the Single group's RT is still 444 ms higher. This is ANCOVA logic — controlling for a potential confound. The fact that FA is non-significant as a covariate tells you that error rate doesn't explain the RT difference between groups.



Target Density GLM — Explaining the RT Paradox

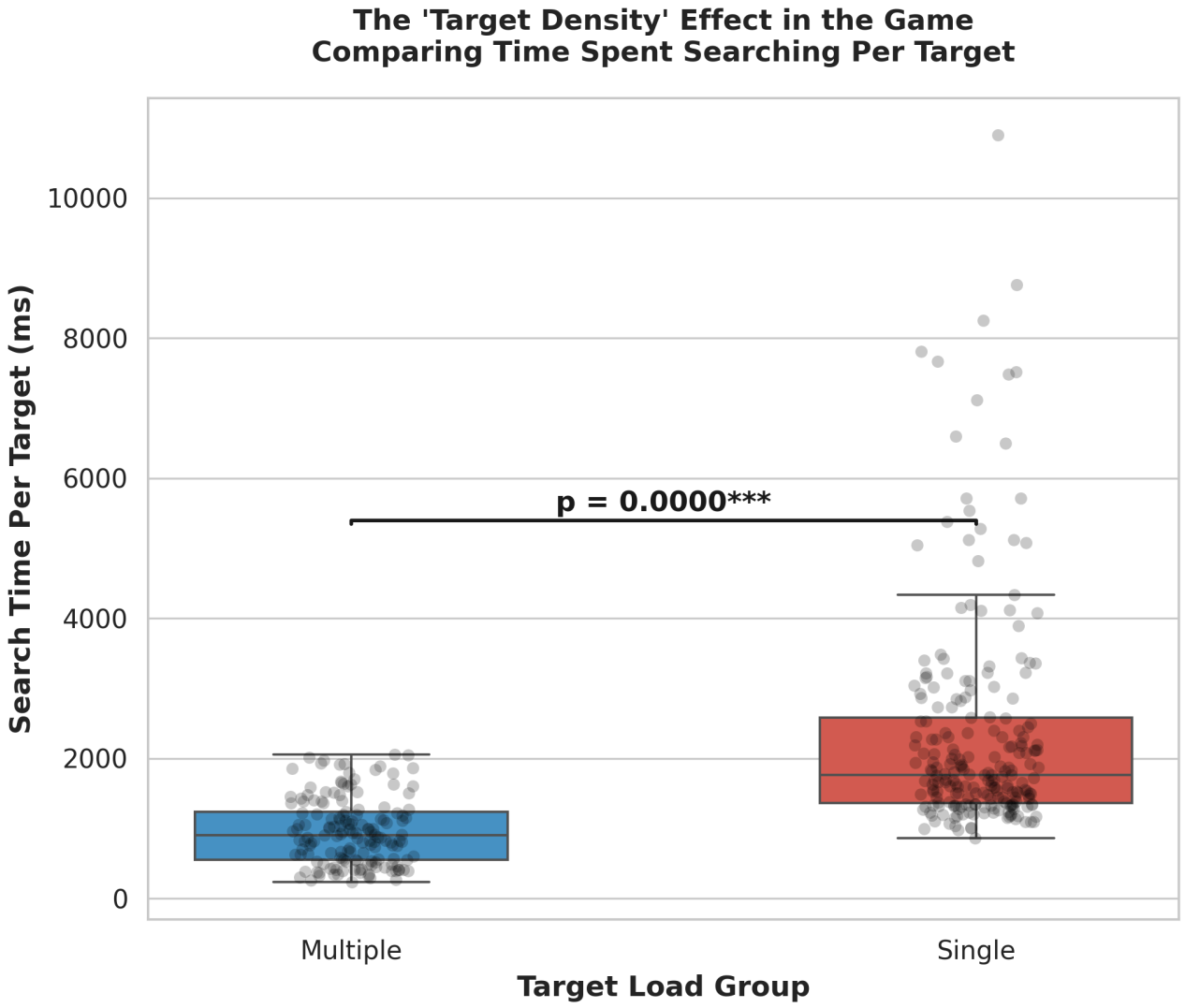
The final GLM explains *why* Multiple group appears faster:

```
Per_target_search_time ~ Group   (Welch t-test equivalent)
```

Per-target search time = total level RT ÷ number of targets on screen.

- Multiple: M = 959 ms per target
- Single: M = 2339 ms per target
- $t = 12.21$, $p < .0001$

The logic: If 3–5 targets are scattered on screen, the probability that your eyes encounter *any* target quickly is much higher than if there's only 1. So Multiple participants' recorded RT (time to first click) is low not because they're better attenders, but because the probability of randomly landing on a target is 3–5× higher.



Summary: Which Model Was Used for What

Purpose	Model	Why This Model
Test the interaction hypothesis	2×2 Mixed ANOVA	Classic, interpretable; handles between + within factors
Confirm effect direction within groups	Paired t-test (within), Independent t-test (between)	Simple, direct; paired removes between-person noise
Account for individual differences	LME	Theoretically ideal for repeated measures; revealed ICC=0
Quantify multiplicative RT cost of gamification	Gamma GLM	Correct distribution for positive, skewed RT data

Purpose	Model	Why This Model
Check if clustering matters	GEE	Sensitivity analysis for LME singularity
Disentangle RT from error rate	ANCOVA (GLM)	Controls for false alarms as covariate
Explain RT paradox	Per-target Welch t-test	Direct comparison of density-corrected metric
Test difficulty scaling	Spearman ρ	Ordinal level variable; monotonic not linear
Test colour guidance (FIT)	Mann-Whitney U	Non-parametric; ICI is heavily skewed
Test serial exhaustion (FIT)	Kruskal-Wallis H	Non-parametric; 4+ rank groups